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Pre-Board Examination 2025-26

Class - X

Subject: Mathematics

Full Mark: 80

General Instructions :

Read the following instructions very carefully and strictly follow them :

- (i) This question paper contains 38 questions. All questions are compulsory.
- (ii) This question paper is divided into five Sections – A, B, C, D and E.
- (iii) In Section A, Questions no. 1 to 18 are Multiple Choice Questions (MCQs) and questions number 19 and 20 are Assertion-Reason based questions of 1 mark each.
- (iv) In Section B, Questions no. 21 to 25 are Very Short Answer (VSA) type questions, carrying 2 marks each.
- (v) In Section C, Questions no. 26 to 31 are Short Answer (SA) type questions, carrying 3 marks each.
- (vi) In Section D, Questions no. 32 to 35 are Long Answer (LA) type questions carrying 5 marks each.
- (vii) In Section E, Questions no. 36 to 38 are case study based questions carrying 4 marks each. Internal choice is provided in 2 marks questions in each case study.
- (viii) There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 2 questions in Section C, 2 questions in Section D and 3 questions in Section E.
- (ix) Take $\pi = \frac{22}{7}$ wherever required, if not stated.
- (x) Use of calculator is not allowed.

SECTION A

This section comprises Multiple Choice Questions (MCQs) of 1 mark each.

1. If 5 times the 5th term of an A.P. is equal to 9 times the 9th term, then its 14th term is :
(A) 5 (B) 9
(C) 0 (D) 14

P.T.O.

2. If the sum and the product of the roots of the quadratic equation $ax^2 + 6x + 4a = 0$ are equal, then 'a' is equal to :

- (A) $\frac{3}{2}$
 (B) $-\frac{3}{2}$
 (C) $\frac{2}{3}$
 (D) $-\frac{2}{3}$

$$\frac{3}{2} = \frac{6}{4}$$

3. If the lines given by $3x + 2ky = 2$ and $2x + 5y = 1$ are parallel, then the value of k is :

- (A) $-\frac{5}{4}$
 (B) $\frac{2}{5}$
 (C) $\frac{15}{4}$
 (D) $\frac{3}{2}$

$$\frac{3}{2} = \frac{2k}{5}$$

$$3 \times 5 = 2 \times \frac{3}{2}$$

$$15 = 3$$

$$\frac{3}{2} = \frac{15}{5}$$

$$\frac{3}{2} = \frac{3}{2}$$

4. If α and β are the zeroes of the polynomial $2x^2 + 5x + 1$, then the value of $\alpha + \beta + 3\alpha\beta$ is :

- (A) -4
 (B) $-\frac{3}{2}$
 (C) 1
 (D) -1

$$\alpha + \beta = -\frac{5}{2}$$

$$\alpha\beta = -\frac{1}{2}$$

$$-\frac{5}{2} + 3 \times -\frac{1}{2}$$

$$-\frac{5}{2} - \frac{3}{2} = -\frac{8}{2} = -4$$

5. If $a = 2^4 \times 3^3$, $b = 2^3 \times 3^2 \times 5$, $c = 3^n \times 5^2$ and

LCM (a, b, c) = $(5^2 \times 3^4 \times 2^4)$, then n is :

- (A) 1
- (B) 2
- (C) 3
- (D) 4

6. In the following frequency distribution :

Height (in cm) :	120 – 125	125 – 130	130 – 135	135 – 140	140 – 145
Number of students :	17	12	13	8	10

the sum of the upper limit of the modal class and the lower limit of the median class is :

- (A) 250
- (B) 255
- (C) 260
- (D) 245

$\frac{8}{10}$
 $\frac{12}{12} =$

7. If the median and mode of a frequency distribution are 26 and 29 respectively, then the mean is :

- (A) 27.5
- (B) 24.5
- (C) 28.4
- (D) 25.8

$3 \times 26 - 29$
 $+ 29 - 29$ 59

8. The probability for a leap year (selected at random) to have 52 Mondays and 53 Sundays is :

- (A) $\frac{1}{366}$
- (B) $\frac{1}{52}$
- (C) $\frac{2}{7}$
- (D) $\frac{1}{7}$

P.T.O.

9. The probability that in a family of three children, there will be at least two boys, is :

- (A) $\frac{1}{8}$
- (B) $\frac{7}{8}$
- (C) $\frac{4}{8}$
- (D) $\frac{6}{8}$

10. A hemispherical bowl is made of steel of thickness 0.30 cm. The inner radius of the bowl is 3 cm. The volume of steel used (in cu cm) is :

- (A) 595.8π
- (B) 5.958π
- (C) 6π
- (D) 59.58π

11. A sector of a circle with central angle 120° and area $\frac{264}{7}$ sq cm is cut from a circle. The radius of the circle (in cm) is :

- (A) 6
- (B) 5
- (C) 7
- (D) 12

12. If $\operatorname{cosec} \theta = \sqrt{10}$, then the value of $\sec \theta$ is :

- | | |
|---------------------------|---------------------------|
| (A) $\frac{3}{\sqrt{10}}$ | (B) $\frac{\sqrt{10}}{3}$ |
| (C) $\frac{1}{\sqrt{10}}$ | (D) $\frac{2}{\sqrt{10}}$ |

13. $\frac{\sin^3 A + \cos^3 A}{\sin A + \cos A} + \sin A \cos A$ on simplification gives :
 (A) 1 (B) 2
 (C) $1 + 2 \sin A \cos A$ (D) 0
14. The angle of depression of a car parked on the road from the top of a 75 m high tower is 30° . The distance of the car from the base of the tower is :
 (A) $75\sqrt{3}$ m (B) $50\sqrt{3}$ m
 (C) $25\sqrt{3}$ m (D) 75 m
15. In two concentric circles with centre O, the radius of the outer circle is 50 cm. Chord AB of the outer circle is tangent to the inner circle at D. If length of AB is 96 cm, then the radius of the inner circle is :
 (A) 14 cm (B) 7 cm
 (C) 24 cm (D) 15 cm
16. In two Δ s ABC and PQR, if $\frac{AB}{QR} = \frac{BC}{QP} = \frac{AC}{PR}$, then
 (A) $\Delta PQR \sim \Delta CAB$ (B) $\Delta PQR \sim \Delta ABC$
 (C) $\Delta PQR \sim \Delta CBA$ (D) $\Delta PQR \sim \Delta BCA$
17. In a ΔABC , a line DE is drawn parallel to BC to intersect AB at D and AC at E. If AD = 2 cm, BD = 3 cm and DE = 4 cm, then the length of BC (in cm) is :
 (A) 6 (B) 10
 (C) $\frac{8}{3}$ (D) $\frac{20}{3}$

18. The points $(-2, -2)$, $(6, -2)$ and $(2, 1)$ are the vertices of :

- (A) a right angled triangle
- ✓(B) an isosceles triangle
- (C) an isosceles right triangle
- (D) a scalene triangle

Questions number 19 and 20 are Assertion and Reason based questions. Two statements are given, one labelled as Assertion (A) and the other is labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
- (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of Assertion (A).
- (C) Assertion (A) is true, but Reason (R) is false.
- (D) Assertion (A) is false, but Reason (R) is true.

19. *Assertion (A)* : Two cubes each with 12 cm edge are joined end to end. The surface area of the resulting cuboid = $2 \times$ (surface area of one cube).

Reason (R) : The surface area of a cuboid = $2(lb + bh + hl)$, where l , b , h respectively are its length, breadth and height.

20. *Assertion (A)* : The eighth term of the A.P.

$$\frac{1}{m}, \frac{1+2m}{m}, \frac{1+4m}{m}, \dots \text{ is } \frac{1+14m}{m}.$$

Reason (R) : The n^{th} term of A.P. $(a_n) = a + (n - 1) d$.

SECTION B

This section comprises Very Short Answer (VSA) type questions of 2 marks each.

21. E and F are points on the sides PQ and PR respectively of a ΔPQR . If $PE = 3.9$ cm, $EQ = 3$ cm, $PF = 3.6$ cm and $PR = 6$ cm, find whether $EF \parallel QR$.

22. (a) Prove that the line segment joining the points of contact of two parallel tangents to a circle passes through its centre.

OR

(b) Two tangents PQ and PR are drawn from an external point P to a circle with centre O. Prove that QORP is a cyclic quadrilateral.

23. If $12 \operatorname{cosec} A = 13$, then find the value of $\frac{2 \sin A - 3 \cos A}{4 \sin A - 9 \cos A}$.

24. The length of the minute hand of a wall clock is 21 cm. Find the area swept by the minute hand in 45 minutes.

25. (a) Find whether each of the following is an irrational number or a rational number.

(i) $(\sqrt{5} - \sqrt{3})^2$

(ii) $(5 + \sqrt{3})(5 - \sqrt{3})$

OR

(b) Find the smallest 4-digit number exactly divisible by 15, 24 and 36.

P.T.O.

SECTION C

This section comprises Short Answer (SA) type questions of 3 marks each.

26. (a) Prove that the parallelogram circumscribing a circle is a rhombus.

OR

- (b) Prove that the tangents drawn at the ends of a diameter of a circle are parallel.

27. Places A and B are 160 km apart on a highway. One car starts from A and another from B at the same time. If the cars travel in the same direction at different speeds, they meet in 4 hours. If they travel towards each other, they meet in 1 hour 36 minutes. What are the speeds of the two cars ?

28. (a) If α and β are the zeroes of the quadratic polynomial $f(x) = 6x^2 + 11x - 10$, find the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$.

OR

- (b) Find the zeroes of the polynomial $2t^2 - 9t - 45$ and verify the relationship between the zeroes and the coefficients of the polynomial.

29. Prove that $\sqrt{5}$ is an irrational number.

30. Prove that $\sqrt{\sec^2 A + \operatorname{cosec}^2 A} = \tan A + \cot A$.

31. The table below shows the daily expenditure on food of 25 households in a locality :

Daily expenditure (in ₹) :	200 – 250	250 – 300	300 – 350	350 – 400	400 – 450
Number of households :	4	5	12	2	2

Find the mean daily expenditure on food.

Stirred childhood

SECTION D

This section comprises Long Answer (LA) type questions of 5 marks each.

32. (a) Some students planned a picnic. The total budget for food was ₹ 500, but 5 of them failed to go and thus the cost of food for each student increased by ₹ 5. How many students attended the picnic?

OR

- (b) Find the value of p if the equation $(2p + 1)x^2 - (7p + 2)x + 7p - 3 = 0$ has real and equal roots.

33. Find the coordinates of the points which divide the line segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts.

34. If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio.

35. (a) A solid is in the form of a right circular cylinder with hemispherical ends. The total height of the solid is 58 cm and the diameter of the cylinder is 28 cm. Find the total surface area of the solid. 5104

OR

- (b) From a solid cylinder of height 36 cm and diameter 14 cm, a conical cavity of radius 7 cm and height 24 cm is drilled out. Find the volume of the remaining solid.

P.T.O.

SECTION E

This section comprises 3 case study based questions of 4 marks each.

Case Study - 1

36. Hari wants to participate in a 200 m race. He can currently run that distance in 51 seconds, and with each day of practice, he hopes to take 2 seconds less than the previous day. He wants to do it in 31 seconds.

Based on the above information, answer the following questions :

- (i) Write the A.P. which represents the above situation.
- (ii) Find the minimum number of days he needs to practice to achieve the goal.
- (iii) (a) Find the expression for the n^{th} term of the A.P.

OR

- (b) If he wants to do it in 21 seconds, how many minimum days will he take ?

Case Study - 2

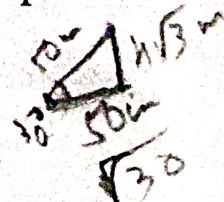
37. A class VI student went to a park and went up the slide to play. The angle of elevation of the slide is 30° , but the base from which the angle of elevation is measured is 50 cm above the ground level and the distance of this point from the bottom of the staircase (which is vertical) is $4\sqrt{3}$ m.

Based on the above information, answer the following questions:

- (i) Write the angle of depression from the top of the slide to its base.
- (ii) (a) Find the height of the staircase.

OR

- (b) Find the length of the slide.
- (iii) Will the angle of elevation increase or decrease if the staircase was made taller ?



Case Study – 3

38. A survey was conducted by the Education Ministry of India to record the teacher–student ratio in various higher secondary schools of India. The following distribution was given by the Ministry :

Number of students/teacher :	15 – 20	20 – 25	25 – 30	30 – 35	35 – 40	40 – 45
Number of Schools :	3	8	9	10	3	2

Based on the above information, answer the following questions :

- (i) Write the modal class. 1
- (ii) Write the median class. 1
- (iii) (a) Find the mode of the data. 2

OR

- (b) Find the median of the data. 2